

Contact

amfkey@gmail.com

www.linkedin.com/in/akey7
(LinkedIn)

scialicia.com (Portfolio)

Top Skills

Data-Driven Insights

Model Development

Research Analysis

Languages

English (Native or Bilingual)

Certifications

Tableau Desktop

Power Query, Power Pivot

Cloud Basics for Data Professionals

Business Charts in Excel

Publications

Database-as-a-Service for Long Tail Science

VizDeck: Self-Organizing Dashboards for Visual Analytics.

VizDeck: a card game metaphor for fast visual data exploration

Alicia Michelle Farley Key

Systems Biology | PhD Candidate | NIH F31 Fellow

Aurora, Colorado, United States

Summary

My work in data analysis and visualization has been featured in ten papers and technical reports, and published in eight biomedical and renewable energy journals. I have had the privilege of collaborating on teams sponsored by the U.S. Department of Energy, the National Renewable Energy Laboratory, and the National Institutes of Health. I am passionate about exploring new datasets and bringing insights to life, regardless of the field!

Top Technical Skills

- Programming: Python, R, SQL

- Data analysis and visualization: Excel, Power BI, Tableau

- Databases: PostgreSQL, MySQL

Experience

University of Colorado Anschutz Medical Campus

PhD Candidate, Structural Biology, Biochemistry, and Biophysics

August 2022 - Present (3 years 6 months)

Aurora, Colorado, United States

Using Excel, R, and machine learning, contributing author to 6 academic publications in the biomedical field listed on the National Institute of Health's PubMed database.

Using Excel and R: machine learning and visualization of data from 13,091 consenting donors in a blood donation study sponsored by the National Heart, Lung, and Blood Institute. Co-first author on "Genetic regulation of carnitine metabolism controls lipid damage repair and aging RBC hemolysis in vivo and in vitro". Figure 1B, 1C, 1F (paywalled, abstract freely available).

Using Excel and R: machine learning and visualization of data from 640 consenting donors in a blood donation study sponsored by the National Heart, Lung, and Blood Institute. Contributing author to "Regulation of kynurenine

metabolism by blood donor genetics and biology impacts red cell hemolysis in vitro and in vivo,” figures 2G and 3A (paywalled, abstract freely available).

Full text copies of publications available upon request.

National Renewable Energy Laboratory

Researcher III

April 2019 - August 2022 (3 years 5 months)

Golden, Colorado

Using Tableau and Python, data analysis and visualization with Tableau and Python for 5 published papers and technical reports with topics relating to renewable energy.

Using Python and Excel, expanded a cost model of wind plant construction costs from an Excel-based model that could only model 1 scenario at a time to a Python model that could model 1000s of power plants at once.

Using Tableau: Wind power plant construction cost modeling. First author of “Scaling trends for balance-of-system costs at land-based wind power plants: Opportunities for innovations in foundation and erection” in Wind Engineering (paywalled, abstract freely available, full-text available upon request).

Using Tableau, Python, Plotly, and Dash, analyzed technologies to recycle decommissioned wind turbine blade. “The Circular Economy Life Cycle Assessment and Visualization Framework: A Multistate Case Study of Wind Blade Circularity in United States,” figures 3, 4, 5 (open access).

Galvanize Inc

Instructor

November 2016 - March 2019 (2 years 5 months)

Boulder, CO

Taught over 100 students as an instructor.

Using Python, created/curated curriculum and taught “Introduction to Data Science” and “Python Fundamentals” part-time evening and weekend courses. Topics included Python, data visualization, basic modeling, linear algebra and statistics to prepare students for later coursework in AI, machine learning, and deep learning, Pandas, NumPy, matplotlib, linear regression and logistic regression, statistics and probability.

Using Excel, SQL, and Tableau, taught the inaugural part-time Data Analytics course at the Boulder Galvanize campus.

Angi

Software Engineer

April 2016 - October 2016 (7 months)

Fort Collins, Colorado, United States

- Developed front end code in JavaScript and back end code in Scala.

ThreatWave

Software Engineer

November 2015 - February 2016 (4 months)

Louisville, Colorado, United States

- Front end development in JavaScript and backend development in Scala.

WYOMING DEPARTMENT OF TRANSPORTATION

Temporary Employee - GIS/ITS

June 2015 - October 2015 (5 months)

Cheyenne, Wyoming, United States

- Developed apps for internal users using Oracle and SQL.

POSSIBLE

Software Engineer

January 2014 - January 2015 (1 year 1 month)

Denver, Colorado, United States

- Developed iOS apps with Objective-C.

UCAR - The University Corporation for Atmospheric Research

Software Engineer/Programmer II

February 2012 - September 2013 (1 year 8 months)

Boulder, Colorado, United States

- JavaScript and Ruby.

University of Washington eScience Institute

Research Engineer

October 2010 - January 2012 (1 year 4 months)

Seattle, Washington, United States

- JavaScript and SQL for data visualization.

Education

University of Colorado Anschutz Medical Campus
Doctor of Philosophy - PhD, Biochemistry · (August 2022 - August 2026)

University of Washington
Master of Science - MS, Chemistry · (2008 - 2010)